Band-1 Hadamard Order 5188 — Miyamoto (Double Paley-I seed H_1296); in SPK via Paley-II (q=2593); Gap 2052 Still OPEN

H_648/H_1296/C_1298/H_5188 PASS; Paley-II H_5188 PASS; 5188 in SPK via Paley-II; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_5156_Miyamoto`; `2026-09-15_Band1_Hadamard_4948_Miyamoto` (in-SPK Paley-II pattern).
Lab. `/workspace/experiments/band1/hadamard_5188_miyamoto/`.
Cutoff. 2026-09-15T18:51:44Z (~2:51PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 648, 1296 and 5188 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Double(Paley-I) seed H_{1296} (H_{648} = Paley-I($q = 647$); H_{1296} = Sylvester-double(H_{648})), and via classical Paley-II ($q = 2593$). Paley-II path for H_{1296} is **FAIL** ($q = 647 \equiv 3 \pmod 4$; needs $\equiv 1$). Paley-I direct for H_{1296} is **SKIP** ($q = 1295 = 5 \cdot 7 \cdot 37$ not prime). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 5188 is **in SPK via Paley-II** (prime $q = 2593$): Paley-I FAIL ($q = 5187 = 3 \cdot 7 \cdot 13 \cdot 19$), Paley-II PASS ($q = 2593$ prime $\equiv 1 \pmod 4$), Kronecker FAIL (4×1297). Same honesty pattern as 4948/4804/4036 (in SPK, not Miyamoto-only). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 5156 on box. This increment ships **order 5188** ($5188 = 4 \cdot 1297$, 1297 prime $\equiv 1 \pmod 4$) via Miyamoto with a **Double(Paley-I) seed** H_{1296}. Symmetric Paley conference C_{1298} over prime 1297; $H_{5188} = \text{Miyamoto}(H_{1296}, C_{1298})$. Classical Paley-II path for order 5188 ($q = 2593$) also verified on box. Unit tests **62/62 PASS**. SPK next gap remains **2052** (OPEN). Order 5188 is **in SPK via Paley-II** (NOT Miyamoto-only). Seeds H_{648}/H_{1296} enter SPK via Double Paley-I. The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 5188; Miyamoto; Paley type I; Double Paley-I; Paley type II; Paley conference; constructive gap 2052; SPK; Band-1; in-SPK via Paley-II; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in ± 1 satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{648} constructed and verified (Paley type I, $q = 647$). 2. H_{1296} constructed and verified (Sylvester double of H_{648}). Paley-II path for H_{1296} **FAIL** ($q = 647 \equiv 3 \pmod 4$). Paley-I direct **SKIP** ($q = 1295 = 5 \cdot 7 \cdot 37$). 3. C_{1298} constructed and verified (Paley conference, $q = 1297$): $CC^\top = 1297I$, zero diagonal. 4. H_{5188} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 5. H_{5188} also constructed via Paley-II ($q = 2593$) and verified. 6. SPK: $5188 \in$ SPK via Paley-II (prime $q = 2593$). Paley-I / Kronecker paths for 5188 **rejected**. Next constructive gap remains **2052** (OPEN). Verified H_{648}/H_{1296} enter SPK as Double Paley-I seeds. Prior 5156 remains Miyamoto-only (excluded from SPK seeds). Prior 4948 remains in SPK via Paley-II ($q = 2473$).

2. Arithmetic and eligibility

Quantity	Value
Target order	$5188 = 4 \cdot 1297$
$q = 1297$	prime, $1297 \equiv 1 \pmod 4 \to$ Miyamoto-eligible
Needed H_{q-1}	H_{1296}
Needed conference	order $1298 = q + 1$
H_{1296} path (used)	Double(Paley-I): $H_{648} = \text{Paley-I}(q = 647)$; $H_{1296} = \text{Sylvester-double}(H_{648})$
H_{1296} Paley-II	**FAIL**: $2(q+1) = 1296 \Rightarrow q = 647 \equiv 3 \pmod 4$ (needs $\equiv 1$)
H_{1296} Paley-I direct	**SKIP**: $q = 1295 = 5 \cdot 7 \cdot 37$ not prime
Conference path	Symmetric Paley over prime 1297
Paley-I for 5188	**REJECTED**: $q' = 5187 = 3 \cdot 7 \cdot 13 \cdot 19$ not prime

Quantity	Value
Paley-II for 5188	**PASS**: $q' = 2593$ prime $\equiv 1 \pmod 4 \rightarrow$ $n = 2(q+1) = 5188$ **in SPK**
Kronecker for 5188	**REJECTED**: only 4×1297; $1297 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{648}	Paley type I, $q = 647$	0	0.3372	**PASS**
H_{1296}	Sylvester double of H_{648}	0	1.0013	**PASS**
H_{1296} Paley-II	$q = 647 \equiv 3 \pmod 4$	—	—	**FAIL** (needs $\equiv 1$)
H_{1296} Paley-I direct	$q = 1295 = 5 \cdot 7 \cdot 37$	—	—	**SKIP** (not prime)
C_{1298}	Paley conference, $q = 1297$	0	1.7173	**PASS**
H_{5188}	Miyamoto(H_{1296}, C_{1298})	0	67.5471	**PASS**
H_{5188} (SPK)	Paley-II, $q = 2593$	0	72.8200	**PASS**

Primary pipeline wall (incl. unit tests + Paley-II + SPK refresh) $\approx$ 147.2864 s. Compressed ± 1 matrices saved as `H648_pm1.npz`, `H1296_pm1.npz`, `C1298_pm1.npz`, `H5188_pm1.npz`, `H5188_paleyII_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible $\leq 2000 \cup$ on-box prior classical/SPK ships (through 5156/4948/4804/4036/3908/… + **1020/1116/1228/1236** Paley-II + **1032/1092/1152** Paley-I + **524/1048/548/1096/564/1128/600/1200/6** Double Paley-I + **4804** Paley-II prime power 7^4 + **4948** Paley-II $q = 2473$; **exclude** Miyamoto-only 5156/4916/4868/4612/4516/4468/4388/4372/4196/4132/4084/3716/3524/3284/3076/3044/2836/2692/2564) $\cup$ verified H_{648}/H_{1296} (Double Paley-I) $\cup$ verified H_{5188} (Paley-II classical) $\cup$ Sylvester $\cup$ Paley I/II (incl. prime powers), closed under Kronecker (limit 5600). Order 5188 is **in SPK via Paley-II** (folded into SPK extras).

Quantity	Value
Known count	1140
5188 in SPK	**True** (via Paley-II q=2593)
1296 in SPK	**True** (Double Paley-I seed)
648 in SPK	**True** (Paley-I seed)
5156 in SPK	**False** (prior Miyamoto-only)
4948 in SPK	**True** (Paley-II $q = 2473$)
Next constructive gap	**2052** (OPEN)

5. Gap 2052 (method triage / non-finding ONLY)

Paley-I, Paley-II, Kronecker, and Miyamoto-with-Paley-conference paths for order 2052 all **FAIL** on-box arithmetic. This is a **non-finding**, not a counterexample. Gap **2052 remains OPEN**. Conjecture **OPEN**.

6. Non-claims

- No full Hadamard conjecture claim.
- `clay_closed = false` forever in this ship.
- GATE-04 collection **LOCKED**; no invented J.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / finished-ToE claims.
- Order 5188 is **in SPK via Paley-II** (NOT a Miyamoto-only bonus).
- Gap 2052 is **not** filled by this ship.

7. Reproduce

```
cd /workspace/experiments/band1/hadamard_5188_miyamoto
python3 verify_hadamard_5188.py
```

Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `H648_pm1.npz`, `H1296_pm1.npz`, `C1298_pm1.npz`, `H5188_pm1.npz`, `H5188_paleyII_pm1.npz`, this paper.

8. Next optional

After $q = 1297$: **q=1321→n=5284** Double(PaleyI half=660) / PaleyI(q=1319), or **q=1361→n=5444** Kronecker(4×340), or method-attack SPK gap **2052**, or Track B privacy / Object 1 meter await. (q=1301→n=5204 has empty build notes this triage.)